

Data Dictionary

Introduction

This document provides a complete description of all variables included in the dataset. It outlines variable names, definitions, data types. This dataset captures job advertisements and their required skills across programming, analytics, cloud platforms, machine learning, data engineering, business intelligence, and soft skills.

variable	Description	Type
ID	Unique ID for the job posting	Integer
Consultant	Person retrieving the job posting	Text/String
DateRetrieved	Date the job ad was retrieved	Date
DatePublished	Original date of job posting	Date
Job_title	Title of the job vacancy	Text/String
Company	Name of hiring company	Text/String
Knowledge_in	Knowledge of the city/region relevant to job	Text/String
City	Location city	Text/String
Educational_qualifications	Educational qualification	Text/String
Currency	currency	Text/String
Min_Salary	minimum salary offerd	Numeric
Max_Salary	maximum salary offerd	Numeric
URL	URL of the job posting	Text/String
Search_Term	Search term used to find the job posting	Text/String

Job_Category	Category/type of the job	Text/String
Experience_Category	Required experience level (Two or less years, More than 2 and less than 5 years, More than 5 and less than 10 years, More than 10 years, Unknown or Not needed)	Text/String
Seniority Level	Seniority level of the job	Text/String
Location	Job location	Text/String
Mode	Work mode (onsite, remote, hybrid)	Text/String
Payment Frequency	Salary payment frequency	Text/String
year	Year of job posting	Numeric
Employment Type	Type of Employment (Part Time/Full Time/Intern)	Text/String

- Below, all skills are grouped by category. Each variable is **binary (0/1)**

Variable	Description
R	Requirement of R programming language
SAS	SAS analytics
SPSS	IBM SPSS statistical tool
Python	Python programming required
Matlab	Matlab scientific computing
Scala	Scala programming
C#	C# programming
MS Word	Microsoft Word required
Ms Excel	Microsoft Excel requirement
Ms Access	Microsoft Access
Ms PowerPoint	MS PowerPoint

Spreadsheets	Spreadsheets requirment
Data_visualization	Data visualization skill
Presentation_Skills	Presentation ability
Communication	Communication skills
BigData	Big data skills
Data_warehouse	Data warehousing
cloud_storage	General cloud storage
Google_Cloud	Google Cloud Platform
AWS	Amazon Web Services
Machine_Learning	General ML skills
Deep Learning	Deep learning
Java	Java programming skills
C++	C++ programming
C	C programming language
Linux/Unix	Operating system handeling
SQL	General SQL proficiency
NoSQL	Non-relational database systems
RDBMS	Relational database systems
Oracle	Oracle DB
MySQL	SQL-based relational databases
PHP	PHP backend programming
web_design_and_development_tools	web_design_and_development_tools
AI	Artificial Intelligence
Natural_Language_Processing(NLP)	NLP requirements
Microsoft Power BI	Microsoft Power BI tool
Data_marketing	Data-driven marketing skills.
Content_Management	Experience with content management systems
Tableau	Tableau data visualization and dashboard skills
Alteryx	Alteryx analytics platform skills
Spark	Apache Spark skills for big data processing
S3	AWS S3 cloud storage experience

Redshift	AWS Redshift data warehouse skills
Javascript	JavaScript programming skills
Kafka	Apache Kafka streaming platform skills
Bash	Bash shell scripting
Hadoop	Hadoop ecosystem skill
Data_Pipelines	Building and managing data pipelines
MPP_Platforms	Experience with massively parallel processing platforms
Qlik	QlikView/QlikSense BI tool skills
Hive	Hive SQL skills for Hadoop querying.
Tensorflow	TensorFlow deep learning framework skills
Map/Reduce	MapReduce programming skills
agile execution	Agile project execution skills
Data_management	Skills in managing and maintaining data systems
pyspark	PySpark skills for big data processing in Python
Data_mining	Skills in extracting insights from data
Data_science	Data science skills in analysis and modeling
Web_Analytic_tools	Experience with web analytics tools
Numerical_Analysis	Skills in numerical and mathematical analysis
Economic	Knowledge of economics
Finance_Knowledge	Finance knowledge
Investment_Knowledge	Investment knowledge
Problem_Solving	Problem-solving abilities
Team_Handling	Experience managing teams
Bayesian	Bayesian statistics knowledge
Optimization	Skills in optimization techniques
Acturial and risk analysis	Actuarial and risk analysis skills.
Adaptability	Ability to adapt to change
Airflow	Apache Airflow orchestration skill
Analytical_Skills	Analytical thinking skills
ANOVA	Skills in Analysis of Variance statistical technique
Apache	skills in Apache ecosystem

API	API development and integration skills
Automation	Automation skills for tasks or processes
Azure	Microsoft Azure cloud platform experience
biostatistics	Biostatistics skill
Canva	Canva graphic design skills
CI/CD tools	Continuous integration/continuous deployment tools
Clinical research	Clinical research knowledge
Cloud computing	Cloud computing skills
Cloud Data Service	Cloud data service skills
Creating Dashboards	Ability to create dashboards
CS/SE Fundamentals	Computer science/software engineering fundamentals
Data Cleaning	Data cleaning and preprocessing skills
Data Engineering Principles	Principles of data engineering
Data Infrastructure	Knowledge of data infrastructure
Data migration	Skills in migrating data
data Security	Knowledge of data security
Data structures	Knowledge of data structures
Data_governance	Data governance knowledge
Data_modeling	Data modeling skills
DataBriks	Databricks platform skills
Decision Making	Decision-making ability
DevOps	DevOps methodology skills
Docker	Docker containerization skills
Documentation	Ability to document work and processes
Epidemiology	Knowledge of epidemiology.
ETL/ELT	Extract-Transform-Load processes
Exploratory Data Analysis (EDA)	Skills in performing exploratory data analysis
dbt	Familiarity with dbt (data build tool) for data transformation workflows
Fine-tuning	Experience fine-tuning machine learning model
Full-Stack Skills	Full-stack development skills, both frontend and backend

Git	Git version control skills
Github	Experience using GitHub for code collaboration
GitLab	Experience using GitLab for code repository and CI/CD
Global Exposure	Experience working in global projects or teams
Healthcare Analytics	Skills in analyzing healthcare datasets
HTML	HTML skills for web development
Hypothesis Testing	Skills in conducting hypothesis testing for statistical inference
Interpretation skills	Ability to interpret data and analytics results effectively
JMP	JMP statistical software skills
Julia	Julia programming language skills for scientific computing.
Jupyter notebooks	Skills using Jupyter notebooks for interactive coding and reporting
Kubernetes	Kubernetes container orchestration skills
Large Scale data	Experience handling large-scale datasets
Leadership	Leadership skills and team management ability
Looker	Looker BI platform skills
LLM	Large Language Model experience
Matplotlib	Python Matplotlib library skills for plotting
Meta data and master data management	Skills in managing metadata and master data
Microsoft Fabric ecosystem	Microsoft Fabric platform skills
MicroStrategy	MicroStrategy BI tool skills
MLOps tools	Skills with ML operationalization tools
MS SQL Sever	Microsoft SQL Server skills
MS_Teams	Microsoft Teams collaboration skill
Multivariate analysis	Skills in multivariate statistical analysis
Network performance monitoring	Skills in monitoring network systems
NLP	Natural Language Processing skills for text data
Numpy	Python NumPy library skills for numerical computation
OOP(OBJECTED- ORIENTED PROGRAMMING)	Knowledge of object-oriented programming concepts

Organizational skills	Skills in organizing tasks, projects, or teams
pandas	Python pandas library skills for data manipulation
PL/SQL	Oracle PL/SQL database programming skills
Policy-level data interpretation	Ability to interpret data for policy-level decisions
potofolio management	Portfolio management skills
Power_Query	Power Query skills for data transformation
Predictive & Modeling Techniques	Skills in predictive analytics and modeling
Project management	Project management skills
PyTorch	PyTorch deep learning framework skills
R Markdown	Ability to create reproducible reports using R Markdown
Report Preparation	Skills in preparing reports
Retrieval-Augmented Generation (RAG)	Skills in RAG for AI/ML tasks
Scikit	Scikit-learn Python library skills
Semantic Kernel	Semantic Kernel framework skills
Snowflake	Snowflake data warehouse skills
StakeholderManagementSkills	Skills in managing stakeholders
Stata	Stata statistical software skill
Statistical analysis	General statistical analysis skills
Time Series Analysis	Skills in time series data analysis
T-SQL	T-SQL programming skills in Microsoft SQL Server
VBA	Visual Basic for Applications skills
Windows	Windows operating system skills
Writing_Skills	Written communication skills
XGBoost	XGBoost algorithm skills for gradient boosting
zoom	Zoom video conferencing skills
BSc_needed	Whether a Bachelor's degree is required
special in Bsc	Specialization required in Bachelor's degree
MSc_needed	Whether a Master's degree is required
PhD_needed	Whether a PhD is required.
English_needed	Whether English proficiency is required.

